LAB1: Coffee Shop

Scenario:

You are a data analyst working for a local coffee shop, “Brewed Awakenings,” which is interested in improving its sales and customer satisfaction. Your manager has asked you to analyze sales data to uncover trends and insights that can help the shop make better business decisions.

Objective:

Analyze sales data from a local coffee shop using NumPy to uncover trends and insights.

Instructions:

```
daily_sales = np.array([150.50, 200.75, 180.00, 220.25, 300.50, 250.00, 275.75])

week2_sales = np.array([160.00, 210.50, 190.25, 230.00, 310.75, 260.50, 280.00])

print("Daily Sales Data for the Week:")

print(daily_sales)
```

1. **Calculate Total Sales:**
    - Use NumPy to calculate the total sales for the week using sum.
2. **Calculate Average Daily Sales:**
    - Find the average daily sales using mean.
3. **Identify Best and Worst Sales Days:**
    - Determine the best and worst sales days using min and max.
4. **Calculate the Standard Deviation:**
    - Analyze the variability in daily sales by calculating the standard deviation.
5. **Sales Comparison with Weekend Data:**
    - Create a new array for sales during a promotional weekend. [400.00, 450.00]
    - Concatenate the daily sales array with the weekend sales array.
6. **Recalculate Total and Average Sales:**
    - Recalculate the total and average sales including the weekend data.
7. **Visualize Sales Trends:**
    - Create an array representing the days of the week and calculate sales trends.

LAB 2: Bookstore

Objective:

Learn how to create and manipulate DataFrames in Pandas to analyze data related to a bookstore's inventory.

Instructions:

1. **Create a DataFrame:**
    - Manually create a DataFrame representing a bookstore's inventory. Include columns such as "Title", "Author", "Genre", "Price", and "Quantity".

```python
# Step 1: Set Up Your Workspace
import pandas as pd

# Step 2: Create a DataFrame
data = {
    "Title": ["Book A", "Book B", "Book C"],
    "Author": ["Author 1", "Author 2", "Author 3"],
    "Genre": ["Fiction", "Non-Fiction", "Fiction"],
    "Price": [15.99, 22.50, 12.99],
    "Quantity": [10, 5, 8]
}
```

2. **View the DataFrame:**
    - Display the DataFrame to understand its structure.
3. **Access Specific Columns:**
    - Select and print the "Title" and "Price" columns from the DataFrame.
4. **Add a New Column:**
    - Create a new column called "Total Value" that calculates the total value of each book in stock (Price * Quantity).
5. **Filter the DataFrame:**
    - Filter the DataFrame to show only the books that are priced over $15.
6. **Sort the DataFrame:**
    - Sort the DataFrame by "Price" in ascending order.
7. **Group by Genre:**
    - Group the DataFrame by the "Genre" column and calculate the total quantity of books for each genre.
8. **Perform Statistical Analysis:**
    - Calculate the average price of the books in the inventory.

LAB 3: Crime Rate

Duration: 60 Minutes

Level of Difficulty: Intermediate

Objective:

- Use Pandas for data manipulation and analysis.
- Create interactive visualizations with Plotly to gain insights into crime patterns.

Dataset Overview:

In this lab, you will work with a dataset containing crime data across different states and districts in Malaysia. You will explore and analyze the data using Pandas, focusing on understanding the distribution and trends of various crime categories over the years.

The dataset contains the following columns:

1. **State**: Name of the state in Malaysia.
2. **District**: District within the state.
3. **Category**: Crime category (e.g., assault).
4. **Type**: Type of crime.
5. **Date**: Date of the report in YYYY-MM-DD format.
6. **Crimes**: Number of reported crimes.

Instructions:

1. **Setup:**
    - Import necessary libraries: pandas, numpy, and plotly.express.
    - Load the provided crime_district.csv file into a DataFrame.

```
import pandas as pd

import numpy as np


# Load the dataset

df = pd.read_csv("crime_district.csv")
```

2. **Data Exploration:**
    - Display the first 5 rows of the DataFrame.
    - Print the summary statistics for numerical columns.
    - Check for missing values and data types of each column.

3. **Data Cleaning:**
   - ~~Handle missing values if present.~~
   - Rename columns to follow a consistent naming convention (e.g., lowercase, no spaces).
   - Remove duplicate rows if any.
   - Remove Malaysia and All rows
4. **Basic Analysis:**
   - Calculate the total number of crimes for each district In.
     1. Hint: Use the groupby() method with sum().
   - Calculate the total number of crimes for each state.
     1. Using Pandas, group the data by state and calculate the total number of crimes reported for each state.

        Hint: Use the groupby() method with sum().
   - Find the top 5 districts with the highest number of crimes.
     1. Find the top 5 districts that have the highest number of reported crimes.

        Hint: Use groupby() on the district column and sort using sort_values().
   - Identify the trend of crime counts over the years for a specific category (e.g., "assault")
     1. Pick a crime category (e.g., "assault") and analyze the trend of reported crimes over the years.

        Hint: Use filtering and groupby() on the date column to calculate yearly totals.
   - Analyze the distribution of crime types for a given year (e.g., 2020).
     1. Analyze the distribution of different crime types for a specific year (e.g., 2020).

        Hint: Use filtering to select the desired year and create a frequency count using groupby().

Lab 4: Crime Rate Visualization with Matplotlib

Duration

60 minutes

Level of Difficulty

Intermediate

Overview

In this lab, you will create visualizations for the analyses conducted in the previous lab using Matplotlib. You will create bar charts, line plots, and pie charts to visualize the total crimes by state, top 5 districts, crime trends, and crime type distribution.

Synopsis

Participants will perform the following tasks:

1. Create a bar chart to visualize the total crimes by state.
2. Create a bar chart for the top 5 districts with the highest crimes.
3. Plot a line chart to show the trend of "assault" crimes over the years.
4. Create a pie chart to display the distribution of crime types for 2020.

Task

Task 1: Bar Chart - Total Crimes by State

Create a bar chart to visualize the total crimes reported in each state.

Task 2: Bar Chart - Top 5 Districts with the Highest Crimes

Create a bar chart to show the number of crimes for the top 5 districts.

Task 3: Line Chart - Crime Trend for Assault Category

Plot a line chart to display the trend of reported "assault" crimes over the years.

Task 4: Pie Chart - Crime Type Distribution for 2020

Create a pie chart to illustrate the distribution of various crime types for the year 2020.

LAB 5: Data Visualization Using Plotly

Duration:30-45 minutes
Level of Difficulty:Intermediate
Overview:
This lab will focus on using Plotly to create interactive data visualizations for crime statistics data. Participants will learn to create bar charts, scatter plots, line charts, and pie charts using Plotly's graphing tools. The objective is to gain hands-on experience in creating visually appealing and interactive charts.

Synopsis:
Participants will use the Pandas DataFrame created earlier (crime_data_cleaned) to generate interactive visualizations with Plotly. They will explore different chart types to analyze crime statistics by state, district, and type. This lab includes four visualization tasks.

Task:

- **Task 1**: Create an interactive bar chart to show the total number of crimes by state.
- **Task 2**: Build an interactive scatter plot to analyze the relationship between states and crime categories.
- **Task 3**: Create an interactive line chart showing the trend of assault crimes over the years.
- **Task 4**: Design an interactive pie chart to show the distribution of crime types for the year 2020.

Task-by-Task Breakdown:

1. **Task 1: Interactive Bar Chart - Total Crimes by State**
   - Uses px.bar() to create a bar chart.
   - States are on the X-axis, and the number of crimes on the Y-axis.
   - Color intensity varies based on crime count.
2. **Task 2: Interactive Heat Map - Crimes by State and Category**
   - Plot an interactive heatmap where the X-axis shows states and the Y-axis shows crime categories.
   - Color intensity represents the number of crimes (darker color = more crimes).
   - Includes hover tooltips for exploring detailed crime data by state and category.
3. **Task 3: Interactive Line Chart - Trend of Assault Crimes Over the Years**
   - Uses px.line() to show how "assault" crimes have changed over time.
   - line=dict(color='firebrick', width=4) customizes the line's appearance.
4. **Task 4: Interactive Pie Chart - Crime Type Distribution for 2020**
   - Visualizes the distribution of crime types for the year 2020.
   - The pie chart uses hole=0.3 to create a donut-style appearance, and textinfo='percent+label' to show percentages and labels inside the chart.

LAB 6: Analyzing Titanic Survival Data with Python

Estimated Time: 30 minutes

Level of Difficulty: Intermediate

Objectives:

- Gain familiarity with data analysis and visualization using Python and Plotly.
- Learn to handle missing values and perform basic statistical analysis on datasets.

Scenario:

The RMS Titanic was a luxurious British passenger liner that sank on April 15, 1912, during its maiden voyage from Southampton to New York City. Dubbed "unsinkable," the Titanic struck an iceberg, leading to its tragic sinking and the loss of over 1,500 lives.

The disaster exposed class disparities, as first-class passengers had priority in lifeboats while many third-class passengers faced barriers to escape. The Titanic tragedy prompted significant changes in maritime safety regulations, including improved lifeboat requirements.

The story of the Titanic serves as a poignant reminder of human ambition, vulnerability, and the importance of safety in travel.

The Titanic dataset contains information about the passengers aboard the ill-fated RMS Titanic, which sank on its maiden voyage in 1912. The dataset includes various features such as:

1. **survived**: Whether the passenger survived (1) or not (0).
2. **pclass**: Passenger class (1st, 2nd, or 3rd).
3. **name**: Passenger name.
4. **sex**: Gender.
5. **age**: Age in years (some missing values).
6. **sibsp**: Number of siblings/spouses aboard.
7. **parch**: Number of parents/children aboard.
8. **ticket**: Ticket number.
9. **fare**: Passenger fare.
10. **cabin**: Cabin number (many missing values).
11. **embarked**: Port of embarkation (C = Cherbourg, Q = Queenstown, S = Southampton).

Task:

1. Opening the Dataset
2. By using *print(data.isnull().sum())*. There are 3 column that have missing values – Age, Cabin, and Embarked
    a. Replace missing values in the age column with the median age.
    b. Replace missing values in the embarked column with the mode.
    c. Replace missing values in the cabin column with 'Unknown' OR using Mode
    d. Create a new variable family_size that combines sibsp (siblings/spouses) and parch (parents/children).

3. Data Analysis and Visualization
    a. Analyze how many passengers survived the Titanic collision by creating a histogram.
    b. Investigate the impact of passenger age on travel class by calculating the average age by passenger class.
    c. Analyze the correlation between passenger class and ticket fares by calculating the average fare by passenger class.
    d. Investigate the average fare by cabin location (first letter of cabin)
    e. Visualize the survival rate by ticket fare.
    f. **Bonus Analysis**: Investigate the relationship between passenger class and survival rates
    g. **Optional**: Explore other relationships in the dataset, such as the effect of family size on survival.